

Riveting Process Intelligence

Head Diameter Prediction via Linear Regression

"Can you predict a 0.4 mm spec window from press parameters alone?"

0.909

Variance Explained

0.081mm

Prediction Error

1,763

Training Cycles

6

Process Variables

OLS Linear Regression

Ridge (L2)

Lasso (L1)

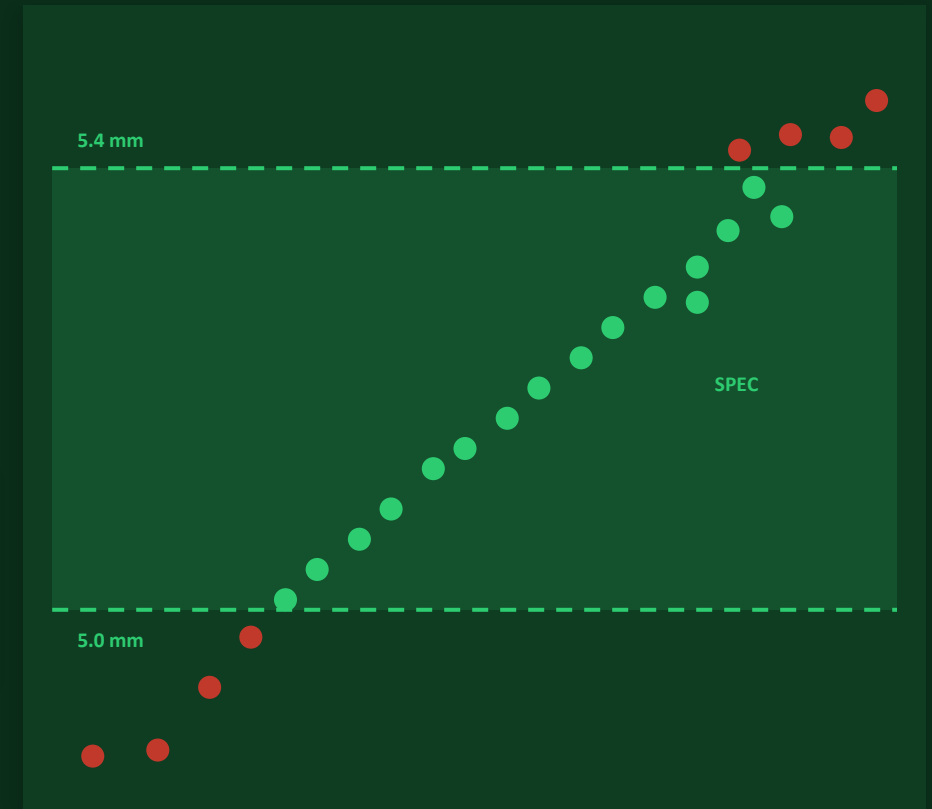
ElasticNet

Automotive · Aerospace · Joining & Assembly · Press Control

LozanoLsa | Operational Excellence · MBB · Machine Learning | GitHub: LozanoLsa

Where $f(x)$ meets Kaizen · Turning Operations into Predictive Systems

A millimeter of error can mean a failed joint or a scrapped panel. The question is: can we see it before the press fires?



Predicted vs Measured · $R^2 = 0.909$



Reactive Quality Loop

By the time the inline CMM confirms a non-conformance, dozens of press cycles have already produced out-of-spec rivets. Inspection confirms failure — it does not prevent it.



Recipe Blindness on Changeover

Switching from a 4 mm to a 5 mm rivet shank without adjusting press force and stroke expands the head beyond spec. The press operator has no predictive reference — only the next CMM report.



50.7% Spec Compliance

Across 1,763 production cycles, barely half of formed rivet heads land within the [5.0–5.4] mm window. The 5 mm rivet family is the worst offender at only 38.8% in-spec.

Insight: The press knows what it is doing — it just cannot tell you in time. That is the gap this model closes.

BEFORE — Reactive Control

- 01 Set Press Recipe**
Operator sets force & stroke from empirical tables or experience.
- 02 Run Production**
Press fires hundreds of cycles. No real-time diameter feedback.
- 03 CMM Sample Inspection**
After-the-fact dimensional check on sample parts.
- 04 Detect Non-Conformance**
CMM reports NG. But 50-100 parts may already be scrapped.
- 05 React & Adjust**
Engineer manually recalibrates. Cycle restarts from scratch.

VS

AFTER — Predictive Control

- 01 Define Process Parameters**
Rivet size, force, stroke, temperature, hold time — 6 inputs.
- 02 Query the Model**
Linear regression predicts head diameter in milliseconds — before the press fires.
- 03 Evaluate Spec Fit**
Model returns predicted mm and margin to nearest spec limit.
- 04 Adjust Recipe if Needed**
Engineer modifies force/stroke until prediction lands inside [5.0–5.4] mm.
- 05 Run with Confidence**
Production runs on a pre-validated recipe. CMM confirms, not discovers.

The model does not replace the engineer — it gives engineering judgment a faster, more quantitative foundation.

Where Each Variable Lives in Your Factory

Variable	Source System	Operational Context	Dataset Profile
press_force_kn	Press PLC	Recorded per cycle by the servo controller	
press_stroke_mm	Press PLC	Ram displacement encoder output	
rivet_diameter_mm	Material Cert / ERP	Nominal shank diameter from incoming inspection	
rivet_length_mm	Material Cert / ERP	Shank length from supplier certificate	
temperature_c	Environmental Sensor	Ambient temp in the press bay at cycle time	
hold_time_ms	Press PLC Timer	Dwell time at full press force	
head_diameter_mm	Inline CMM Station	TARGET — measured formed head diameter	
In real operations, this dataset does not exist in one place. PLC logs, CMM exports, and ERP certificates must be joined — and that integration is 80% of the engineering work.			
Rows		1,763	
Features		6	
Train		1,410	
Test		300	
Spec Lo		5.0 mm	
Spec Hi		5.4 mm	
Spec OK		50.7%	
Rivet Sizes		3/4/5 mm	

Most manufacturing datasets do not arrive clean — they arrive in three different systems with three different timestamps.

"A model that an engineer cannot interrogate is a model an engineer will not trust."

01 Coefficients Speak Engineering

Every coefficient is in physical units — mm per kN, mm per mm of stroke. No black box. The engineer can challenge, validate, and trust each number.

02 Additive Physics Match Reality

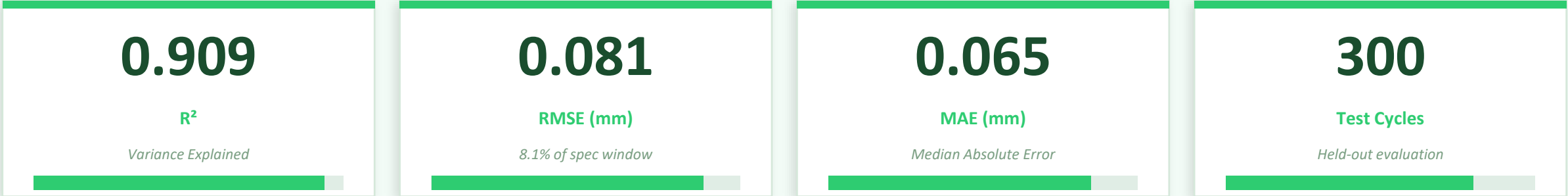
Riveting head formation is a first-order physical process. More force → larger head. Larger rivet → larger head. Linear assumptions hold at 90.9% explained variance.

03 Interpretable for Process Control

The 2D response surface (force × stroke) lets engineers read optimal operating zones directly — making the model a process design tool, not just a predictor.

04 Regularisation Validates Simplicity

Ridge, Lasso, and ElasticNet all converge to the same performance as OLS. When regularisation adds nothing, the signal is real and the model is right.



Feature Coefficients — Engineering Levers

rivet_diameter_mm	+0.186 mm/mm	Dominant driver — more shank material → bigger head
press_stroke_mm	+0.075 mm/mm	Primary controllable lever — adjust for recipe changes
press_force_kn	+0.019 mm/kN	Secondary lever — fine-tune force within recipe
rivet_length_mm	+0.004 mm/mm	Small but consistent across high-volume runs
temperature_c	+0.003 mm/°C	Thermal expansion contributes at scale
hold_time_ms	+0.0002 mm/ms	Minimal — creep effect at extreme dwell times only

Model Validation

Baseline R²

-0.011

Naïve mean model fails

OLS R²

0.909

90.9% variance explained

RMSE vs spec

8.1%

Of the 1.0 mm window

OLS vs Ridge

Equal

Regularisation adds nothing

All coefs

Positive

Physics confirmed

When regularisation does not improve a model, the features are honest — every variable carries genuine signal.

01	Process Engineer Pre-validate Recipe Before Changeover When switching rivet sizes, query the model with the new shank diameter before touching the press. It will indicate whether the current recipe stays in spec — or by how much it will miss.	IMPACT Eliminates first-article scrap on rivet changeovers
02	Quality Engineer Replace Reactive with Predictive CMM Scheduling Prioritise CMM checks for press cycles where predicted diameter falls within 0.05 mm of a spec limit. Focus inspection resources where risk is actually highest.	IMPACT 30–50% reduction in unnecessary CMM measurements
03	Manufacturing Engineer Use the 2D Response Surface as a Process Design Map The force × stroke response surface defines the feasible operating region for any rivet size. Engineers can read optimal recipe zones directly — without physical trials.	IMPACT Faster recipe development for new joint configurations
04	OPEX / Lean Team Quantify the Cost of Temperature and Hold Time Drift Temperature (+0.003 mm/°C) and hold time effects compound across thousands of daily cycles. Use the model to estimate cumulative spec drift during summer ramps.	IMPACT Supports SPC control limits with a physical model basis
05	ML Practitioner Extend to Bayesian Regression for Uncertainty Quantification OLS gives a point estimate: 5.236 mm. It does not say 'with 95% confidence, between 5.174 and 5.298.' A Bayesian or conformal extension adds decision-grade intervals.	IMPACT Enables risk-based process control decisions

Two identical press recipes — one rivet size change. The model predicts the outcome before the first cycle runs.

Scenario A — Standard Operation

Rivet Diameter	4.0 mm
Press Force	42 kN
Press Stroke	4.5 mm
Hold Time	120 ms
Temperature	24 °C
Rivet Length	12.0 mm

Predicted: 5.236 mm

Spec [5.0–5.4] | Margin: +0.164 mm

OK

vs

Scenario B — Unannounced Batch Change

Rivet Diameter	5.0 mm ← CHANGED
Press Force	42 kN (unchanged)
Press Stroke	4.5 mm (unchanged)
Hold Time	120 ms
Temperature	24 °C
Rivet Length	12.0 mm

Predicted: 5.419 mm

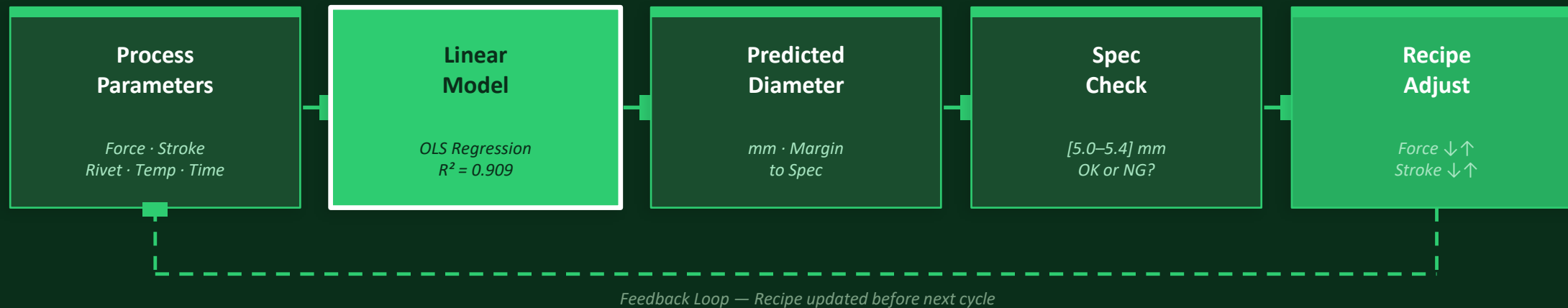
Spec [5.0–5.4] | Margin: –0.019 mm

NG

Scenario C (model-guided adjustment): Force 32 kN + Stroke 3.6 mm → Predicted 5.147 mm → OK (+0.147 mm margin)

Process Intelligence Loop

The model in its natural habitat — not a file, but a feedback system



R^2

0.909

Six process variables explain 90.9% of head diameter variance — physics and regression agree.

0.081

mm RMSE

Prediction error is 8.1% of the spec window — smaller than the measurement system resolution.

> 50k

cycles/yr

At 200 rivets per vehicle and 250 vehicles/day, every percentage of spec improvement is revenue.

A 0.4 mm specification window is not a constraint to inspect around — it is a boundary to predict through.

The press fires. The measurement follows. The model predicts it first.

Clone it. Challenge it. Take it further.